

ROBOT

**(V)KR 150, 180, 210;
(V)KR 150, 180, 210, 240, 270-2**

Maintenance

Manipulator without Wrist

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We have checked the content of this documentation for conformity with the hardware and software described. Nevertheless, discrepancies cannot be precluded, for which reason we are not able to guarantee total conformity. The information in this documentation is checked on a regular basis, however, and necessary corrections will be incorporated in subsequent editions. Subject to technical alterations without an effect on the function.

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Valid for	(V)KR 150	(V)KR 150-2
	(V)KR 150 L 130	(V)KR 150 L 130-2
	(V)KR 150 L 110	(V)KR 150 L 110-2
	(V)KR 180	(V)KR 180-2
	(V)KR 180 L 150	(V)KR 180 L 150-2
	(V)KR 180 L 130	(V)KR 180 L 130-2
	(V)KR 210	(V)KR 210-2
	(V)KR 210 L 180	(V)KR 210 L 180-2
	(V)KR 210 L 150	(V)KR 210 L 150-2
		(V)KR 240-2
		(V)KR 240 L 210-2
		(V)KR 240 L 180-2
		(V)KR 270-2

1 General



This description applies analogously to all of the robots listed above, regardless of the robot variant or model shown in the illustrations.



The Doc. Module "Safety, General" is to be observed!

Maintenance and cleaning work on CR-variant robots may only be conducted outside cleanrooms. Instructions of the cleanroom user must be observed. Cleaning operations are to be carried out with cleaning agents suitable for use in the cleanroom.

Before any maintenance work is started, any attached tools or additional equipment that would hinder maintenance must be dismantled.

Explanation of the symbols used in connection with maintenance intervals in the "Lubrication" and "Other maintenance" tables below.

- means: maintenance interval
- means: recommended maintenance

Lubrication is performed either at the specified maintenance intervals or every 5 years after commissioning by the customer - whichever is reached first.

With maintenance intervals of 20 000 hours, the first maintenance (oil change) is performed either after 20 000 operating hours or 5 years after commissioning by the customer - whichever is reached first.

If the robot is fitted with an energy supply system (option) A 1 – A 2 or A 1 – A 5, additional maintenance work is recommended.



It is imperative that lubrication is carried out every 5 years, as prescribed (oil ageing).



The maintenance intervals given in the tables are valid for the operating conditions specified in the robot Doc. Module "Technical Data". In case of variations from normal conditions (e.g. increased dust or water content in the environment of the robot, abnormal temperatures), the KUKA Roboter GmbH must be consulted.

Explanation of symbols:

Maintenance point



Maintenance point, visible on drawing



Maintenance point, hidden on drawing



Oil change, to be repeated at specific intervals; with indication of oil capacity



Lubricate with grease gun



Lubricate with brush



Tighten screw (nut)



Check workpiece

The description of maintenance operations is subdivided into job steps with numbers in brackets appearing before them. The text which immediately follows these steps must also be read if it is specially marked by a **warning triangle** or either of the **hand symbols**. This is because many of these marked texts refer to the preceding job step.

**Example:**

(1) Pour specified amount of oil into the filler hole at (10).

**Only KUKA-approved lubricants may be used.**

Some of the specially marked texts refer exclusively to everything that follows – until the instruction is expressly revoked or the work is completed at the end of a section.

**Example:**

Turn main switch on the robot control cabinet to “OFF” and secure it with a padlock to prevent unauthorized persons from switching it on again.

2 Lubrication work



Section 1 of this Doc. Module is to be observed!



Turn main switch on the robot control cabinet to “OFF” and secure it with a padlock to prevent unauthorized persons from switching it on again.



Utmost caution must be exercised when handling cable grease. The use of protective clothing (at least protective gloves, for example) is strongly recommended.
All accident prevention regulations and Section 5 of this Doc. Module must be observed.



Store used oil and grease in accordance with regulations and dispose of them with minimum environmental impact.

• “Lubrication” table

Maintenance point (see Fig. 1)	Maintenance interval (hours)		Activity Lubricant Manufacturer Amount KUKA Art. no.	Remarks
	10000	20000		
① Gear with bearings A 1 ¹⁾		●	Oil change Optigear Synthetic RO 150 Optimol, approx. 6.7 l ³⁾ KUKA Art. no. 00-144-898	Change oil as described in Section 2.1.1. Lubricate involute toothing, see Section 2.1.2
② Bearings of counter- balancing system on link arm ²⁾	●		Lubricate LGEP2 grease SFK, approx. 10 g per nipple KUKA Art. no. 00-108-474	Pump grease into nipples until clean grease emerges. Remove old grease.
③ Gear with bearings A 2		●	Oil change Optigear Synthetic RO 150 Optimol, approx. 3.0 l ³⁾ KUKA Art. no. 00-144-898	Change oil as described in Section 2.1.3.
④ Gear with bearings A 3		●	Oil change Optigear Synthetic RO 150 Optimol, approx. 1.9 l ³⁾ KUKA Art. no. 00-144-898	Change oil as described in Section 2.1.4.
⑤ Energy supply system (For optional equipment) A 1 - A 2 A 1 - A 3 A 1 - A 5		□	Check, lubrication Optitemp RB1 Optimol, approx. 1000 g KUKA Art. no. 00-101-456	Verify externally the sliding quality of the flexible tube A 1; if necessary clean the flexible tube and the sliding surface. Apply grease with a cloth. If the sliding quality of the cables and hoses in the flexible tube is no longer sufficient, inform the KUKA Service department and have them re-lubricated.

● normal maintenance interval

□ recommended maintenance

1) Only applicable for robots built before 07/2007: in the case of press linking and highly dynamic handling applications, the connection (involute toothing) between the motor and the gear unit of axis 1 must also be inspected after every 7500 hours of operation and re-greased on both sides with Microlube GL 261 from Klüber Lubrication (KUKA Art. No. 00-135-463) if required.

2) In the case of frequently recurring, short-distance movements about axis 2, the maintenance interval is 3 000 hours.

3) The oil quantities specified are the actual amounts of oil in the gear unit at first filling.

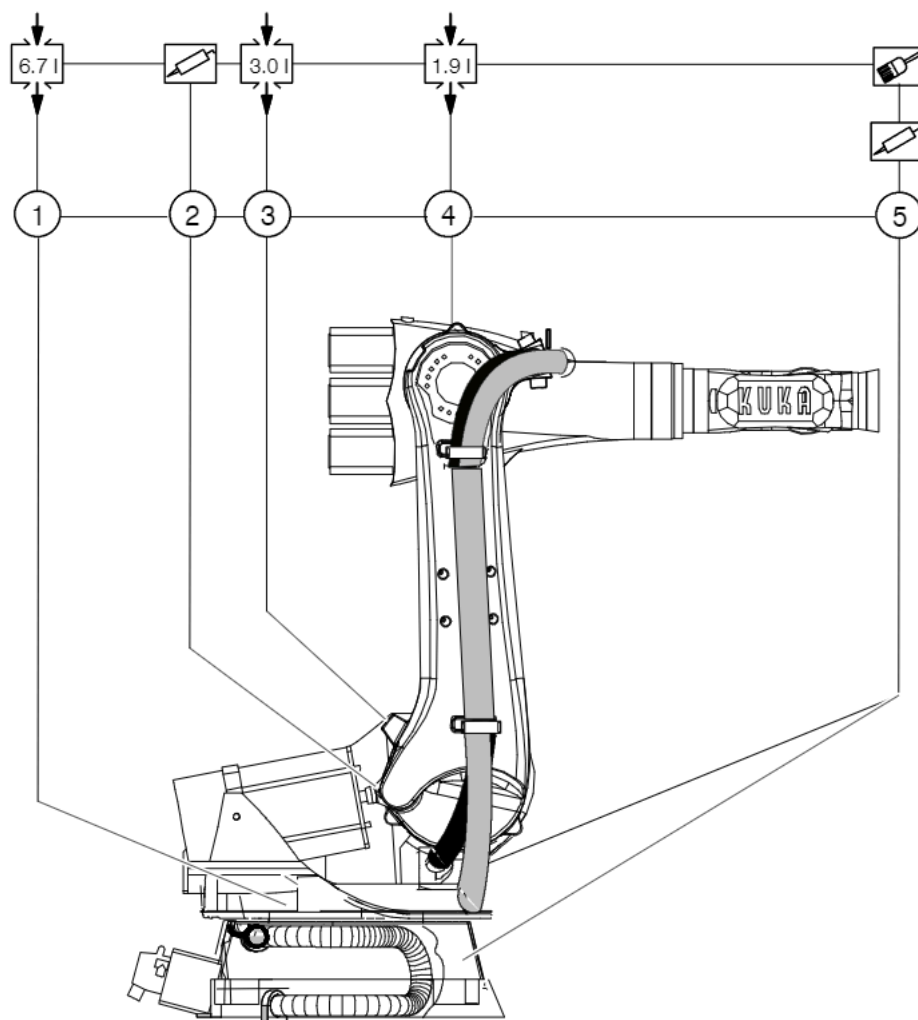


Fig. 1 Lubrication



When draining the oil, it is important to remember that the quantity drained is dependent on time and temperature. The quantity of oil drained must be measured, as only this quantity may be used when refilling.

If less than 70% of the specified oil quantity flows out, flush the gear unit with the measured quantity of drained oil once, then pour in the amount of oil that was drained. During the flushing procedure, move the axis at jog velocity throughout the entire axis range.

2.1 Changing the oil on main axis gear units



Section 2 of this Doc. Module is to be observed!



The oil is to be changed only at operating temperature.



If the oil change is carried out immediately after the robot has stopped operating, the oil temperature is liable to be high, in which case appropriate safety measures must be taken.

2.1.1 Changing the oil on axis 1 gear unit



Section 2.1 of this Doc. Module is to be observed!

• Draining the oil

- (1) Remove four M8x12 Allen screws (Fig. 2/4) including lock washers, take off cover (5) and pull out drain hose (6).
- (2) Unscrew sealing cap (7) on drain hose (6) and place a suitable receptacle under the drain hole.
- (3) Remove screw plug (3) and catch oil as it drains out.



The oil takes about 15 minutes to drain completely.



Store used oil in accordance with regulations and dispose of it with minimum environmental impact.

- (4) Screw sealing cap (7) onto drain hose (6) and tighten it.

• Filling with oil

- (1) Pour specified amount of oil into filler hole of screw plug (3).



Only KUKA-approved lubricants may be used.



Use a hose (2) approx. 1 m long with a straight M33x2 union (8) together with a funnel (1) which fits into the free end of the hose (2).

- (2) Insert and tighten screw plug (3).

- (3) Check sealing cap (7) for leaks.
- (4) Mount cover (5) and fasten it with four M8x12 Allen screws (4) and lock washers.
- (5) Connect any peripheral supply lines required.

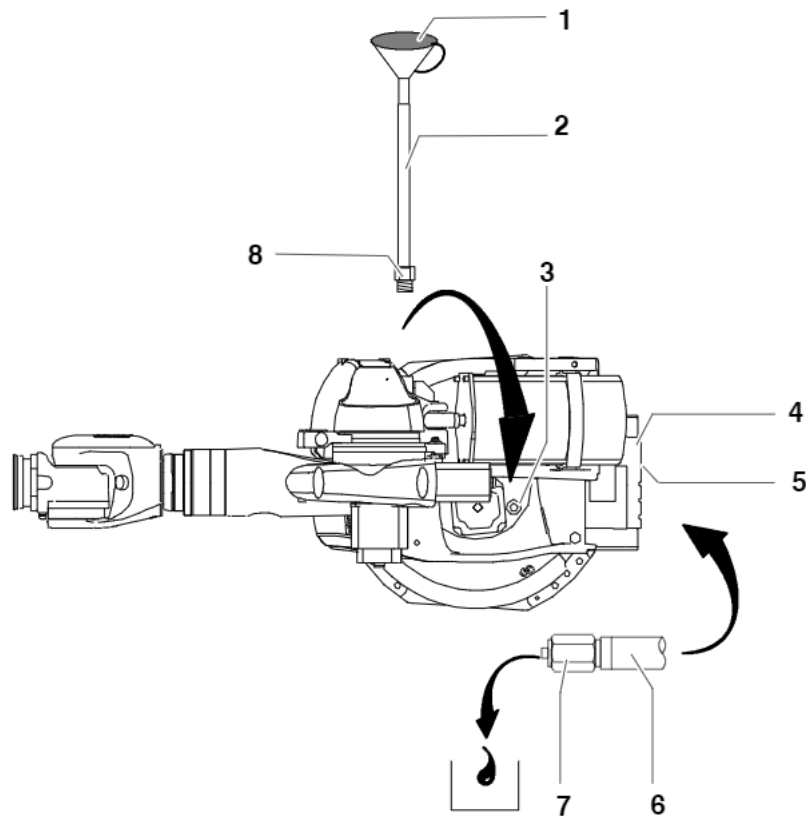


Fig. 2 Changing the oil on axis 1

2.1.2 Lubricating the involute toothing of axis 1



Chapter 1 of this Doc. Module is to be observed!



If AC servomotor A 1 is to be removed from or installed on an operable robot, the main switch on the control cabinet must be turned to "OFF" and secured with a padlock to prevent unauthorized persons from switching it on again.

- Removal



If AC servomotor A 1 is removed immediately after the robot has stopped operating, it is liable to have an elevated surface temperature.

- (1) **For robots in an inclined position:** Secure axis 1 mechanically against rotational movement.



If the robot is installed in an inclined position, the operator must ensure that the robot is not able to move independently during or after removal of AC servomotor A 1.

- (2) Release and unplug connectors XM1 and XP1 at the sockets (Fig. 3/1, 2).
- (3) Unscrew four M12x25-8.8 Allen screws (4).
- (4) Release and lift out AC servomotor A 1 (3).

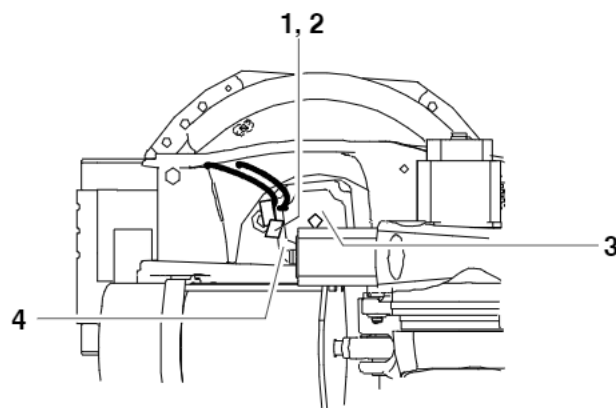


Fig. 3 Removal, installation of AC servomotor A 1



When removing AC servomotor A 1, care must be taken to avoid any injury by crushing!



AC servomotor A 1 must not be tilted while it is being lifted out.



If AC servomotor A 1 is not to be reinstalled, it must be protected against corrosion before being put into storage.

• Inspection and lubrication

- (1) Inspect the toothing of the AC servomotor (Fig. 3/3) and gear unit for fouling (e.g. flash rust).
- (2) Clean the toothing of the AC servomotor (Fig. 3/3) and gear unit.
- (3) Apply a thin but continuous coat of Microlube GL 261 grease to the toothing of the AC servomotor (Fig. 3/3) and gear unit before installation.



If the toothing exhibits irregularities, consult KUKA Service.

• Installation



When installing the AC servomotor it must be ensured that the toothing of both motor and gear unit is OK.

- (1) Remove all protective coatings and oil from new AC servomotor A 1, where applicable.
- (2) Clean toothing of AC servomotor (Fig. 3/3) and gear unit before installation and apply a thin but continuous coat of Microlube GL 261 grease.
- (3) Clean mounting surface for AC servomotor A 1 on gear unit.
- (4) Install AC servomotor A 1 (3).



When installing AC servomotor A 1, care must be taken to avoid any injury by crushing!



AC servomotor A 1 must not be tilted while it is being installed.



The sockets XM1 (1) and XP1 (2) must be positioned as shown in Fig. 3. Insertion of AC servomotor A 1 can be facilitated by turning it gently about its rotational axis.

- (5) Insert four M12x25-8.8 Allen screws (4).
- (6) Tighten the four hexagonal bolts (4) with a torque wrench, increasing the tightening torque M_A in diagonally opposite sequence in several stages to the specified value ($M_A = 78 \text{ Nm}$).
- (7) Plug connectors XM1 and XP1 into the sockets (1, 2).
- (8) **For floor-mounted robots in an inclined position:** Remove safeguards against the robot turning about rotational axis 1.
- (9) Carry out zero adjustment (see Operating Handbook, KR C2, Chapter "Start-up", Section Robot "Mastering/Unmastering").

2.1.3 Changing the oil on axis 2 gear unit



Section 2.1 of this Doc. Module is to be observed!

- **Drain oil**

- (1) Remove sealing cap (Fig. 4/4) and place a suitable receptacle (6) under the drain hole.
- (2) Remove screw plug (3) and catch oil as it drains out.



The oil takes about 15 minutes to drain completely.



Store used oil in accordance with regulations and dispose of it with minimum environmental impact.

- (3) Replace and tighten sealing cap (4) (tightening torque $M_A = 40 \text{ Nm}$).

- **Fill with oil**

- (1) Pour specified amount of oil into the filler hole at (3).



Only KUKA-approved lubricants may be used.



It is easier to pour in the oil if a hose (2) with an M22x1.5 connection (5) and a funnel (1) are used for this purpose.

- (2) Insert and tighten screw plug (3) ($M_A = 40 \text{ Nm}$).
- (3) Check sealing cap (4) for leaks.

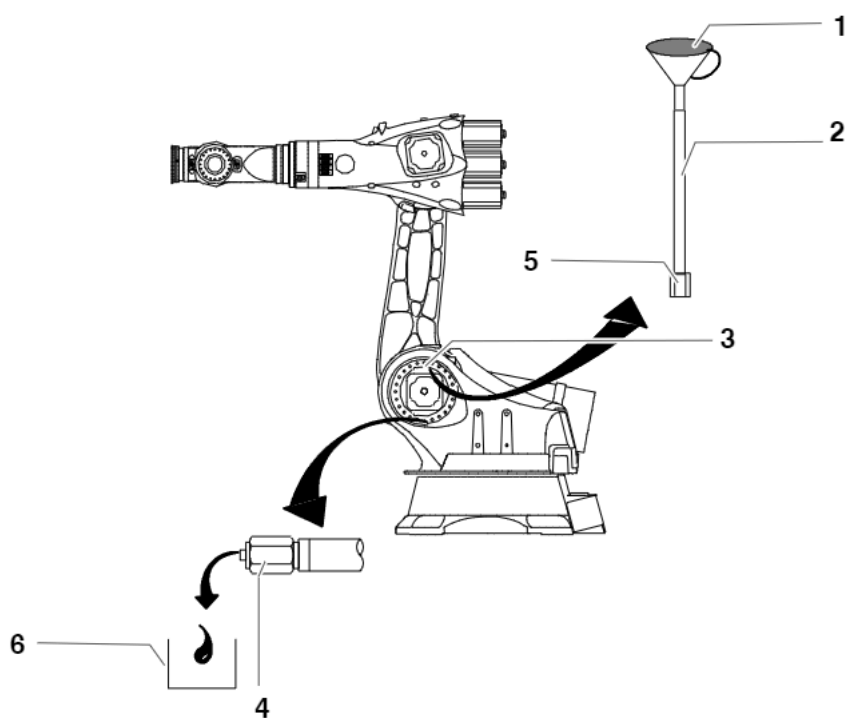


Fig. 4 Changing the oil on axis 2

2.1.4 Changing the oil on axis 3 gear unit



Section 2.1 of this Doc. Module is to be observed!

- Drain oil



Before performing the next step, it must be ensured that it is not possible for anyone to be injured within the range of the slowly turning robot. The robot may only be moved at jog speed, with all applicable safety rules and regulations being observed.

- (1) Put robot into operation and move axis 3 until screw plugs (Fig. 5/2, 6) are aligned vertically.



Turn main switch on the robot control cabinet to “OFF” and secure it with a padlock to prevent unauthorized persons from switching it on again.

- (2) Remove screw plug (6) and place a suitable receptacle (5) under the drain hole.



It is easier to drain the oil if a hose (4) with an M18x1.5 connecting nipple (3) is used.

- (3) Remove screw plug (2) and catch oil as it drains out.



The oil takes about 10 minutes to drain completely.



Store used oil in accordance with regulations and dispose of it with minimum environmental impact.

- (4) Insert and tighten screw plug (6) (tightening torque $M_A = 10 \text{ Nm}$).

- **Fill with oil**

- (1) Pour specified amount of oil into the filler hole at (2).



Only KUKA-approved lubricants may be used.



It is easier to pour in the oil if a funnel is used for this purpose.

- (2) Insert and tighten screw plug (2) ($M_A = 40 \text{ Nm}$).
- (3) Check screw plugs (2, 6) for leaks.

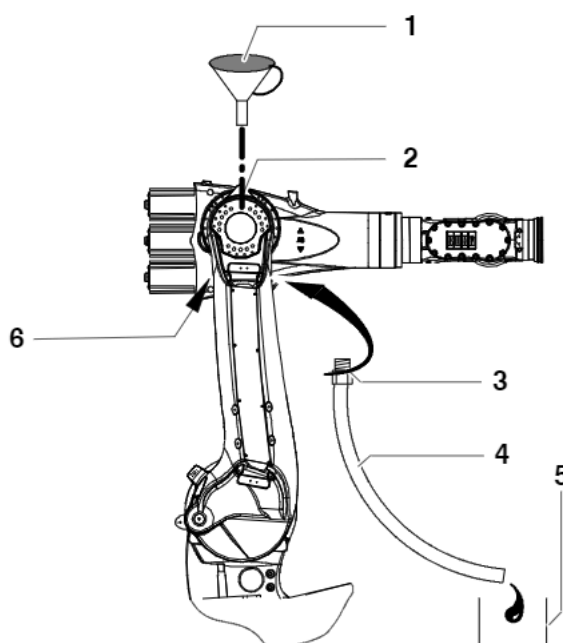


Fig. 5 Changing the oil on axis 3

3 Other maintenance



Section 1 of this Doc. Module is to be observed!



Turn main switch on the robot control cabinet to “OFF” and secure it with a padlock to prevent unauthorized persons from switching it on again.

• “Other maintenance” table

Maintenance point (see Fig. 6)	Maintenance interval (hours)		Activity	Remarks
	10 000	15 000		
① Robot installation	●		Check tightening torques for bolts and anchor nuts	See robot Doc. Module “Installation, Connection, Exchange”.
② Universal shafts	☐ 1)	☐ 2)	Check universal shafts for play. After initial inspection, repeat check annually.	In the case of axial play in the universal joint, exchange the universal shaft.

● Normal maintenance interval

☐ Recommended maintenance

1) For universal shafts in robots built before 08/2004, the interval for the initial inspection is 10,000 operating hours, but no more than 4 years.

2) For universal shafts in robots built in or after 08/2004, the interval for the initial inspection is 15,000 operating hours, but no more than 6 years.



If the robot is fitted with the ProtectionPlus option, additional maintenance work must be carried out (see doc. module Ro/Op/01/33.278).

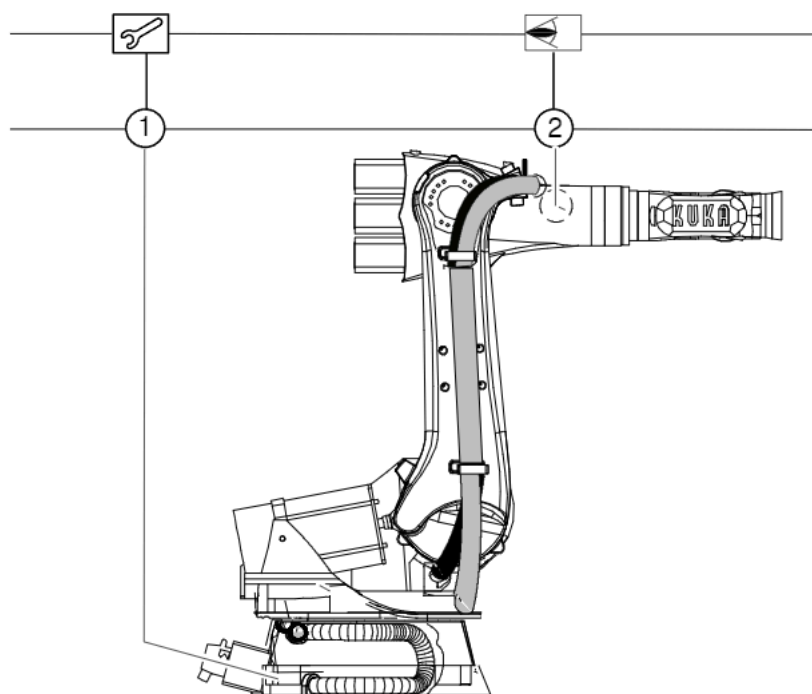


Fig. 6 Other maintenance

4 Cleaning and preventive maintenance



Turn main switch on the robot control cabinet to “OFF” and secure it with a padlock to prevent unauthorized persons from switching it on again. This also applies to any cleaning work, which though not performed directly on the robot is carried out within or close to its working range.

Cleaning and preventive maintenance work is to be carried out in accordance with the following instructions:

- Clean robot with a cloth soaked with washing agent.
- Clean cables, plastic parts and hoses with solvent-free washing agents.



The manufacturer's instructions are to be observed when using washing agents.

- Remove any lubricant that has escaped using washing agent. If a large amount of lubricant has emerged, trace and remedy the cause.



It must be ensured that no washing fluid enters bearings, seals, electrical equipment or the mechanical counterbalancing system.



If washing agent does enter bearings, these must be re-greased or re-oiled.

- Remove any corrosion and – where permissible – protect the affected areas with paint, grease or oil.
- Apply a thin coat of oil to bare metal parts.
- Replace damaged, illegible or missing inscriptions, labels and plates.



Do not use compressed air to clean the system, as this could cause dirt to penetrate and damage seals, bearings and electrical components. Used cleaning agents and lubricants must be stored and disposed of in accordance with all pertinent regulations and with minimum environmental impact.

5 Safety precautions when handling lubricants

The information contained in the safety data sheet according to 91/155 EEC must be observed when handling lubricants. An excerpt from this data sheet is included in the robot Doc. Module "Consumables, Safety Data Sheet".

The following safety procedures must be observed at all times:

- Avoid prolonged intensive skin contact; wear protective gloves and aprons if necessary (especially when working with cable grease).



All accident prevention regulations must be observed when handling lubricants.

- If lubricant is likely to come into contact with the skin, e.g. with the hands, suitable barrier creams must be applied before work is started.
- On completion of work and before meal breaks (before any intake of food), clean oil-contaminated skin thoroughly with water and skin-friendly cleansing agents or soap. After cleaning, replace lost skin grease with appropriate skin creams containing grease.
- Change oil-soaked clothing immediately. Do not carry oily rags or any cloths soaked with solvents, mineral oil mixtures or cable grease in your pockets.
- Avoid breathing in oil spray or vapors wherever possible.